# Introduction of Computer Networks

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# NETWORKS AFFECT OUR LIVES

**1 Networks Connect Us**

In today's world, through the use of networks, we are connected like never before. People with ideas can communicate instantly with others to make those ideas a reality. News events and discoveries are known worldwide in seconds. Individuals can even connect and play games with friends separated by oceans and continents

**2 No Boundaries**

The creation of online communities for the exchange of ideas and information has the potential to increase productivity opportunities across the globe. The creation of the cloud lets us store documents and pictures and access them anywhere, anytime. So whether we are on a train, in a park, or standing on top of a mountain, we can seamlessly access our data and applications on any device.

# NETWORK COMPONENTS

**End Devices :** PC, Laptop, Mobile
**Connecting Devices:** Hubs, Switches, Routers
**Transmission Media:** Wired, Wireless Media
**Software:** Operating Systems for PCs and Connecting Devices

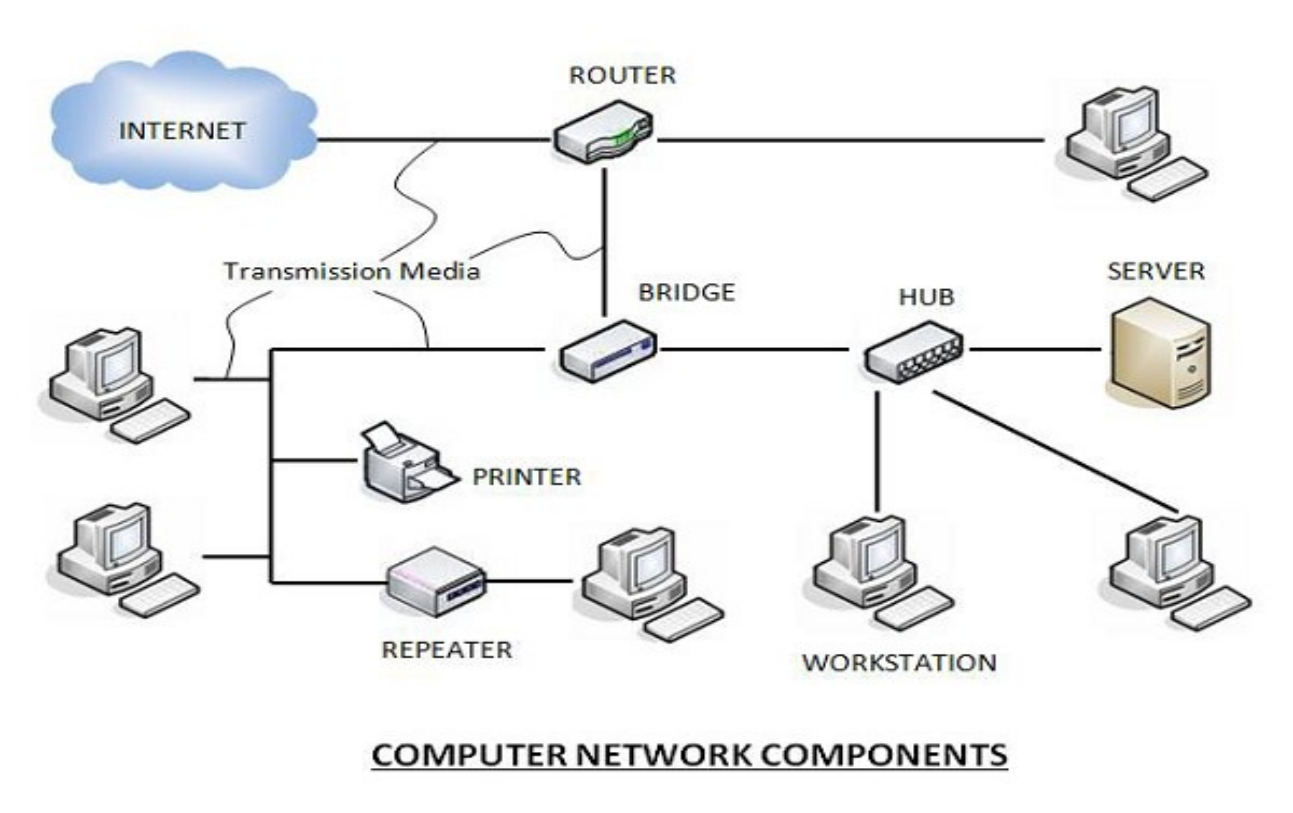



**Figure 1** Shows the network components in detail.

# NETWORK COMPONENTS

## 1 Host

To be a part of a global online community, your computer, tablet, or smartphone must first be connected to a network. That network must be connected to the internet. This topic discusses the parts of a network.

**Host:** All computers that are connected to a network and participate directly in network communication are classified as hosts. Hosts can be called end devices. Some hosts are also called clients. However, the term hosts specifically refers to devices on the network that are assigned a number for communication purposes. This number identifies the host within a particular network.

**IP**: This number is called the Internet Protocol (IP) address. An IP address identifies the host and the network to which the host is attached.

**Server**: Servers are computers with software that allow them to provide information, like email or web pages, to other end devices on the network. Each service requires separate server software. For example, a server requires web server software to provide web services to the network. A computer with server software can provide services simultaneously to many different clients.

As mentioned, clients are a type of host. Clients have software for requesting and displaying the information obtained from the server, as shown in the figure.

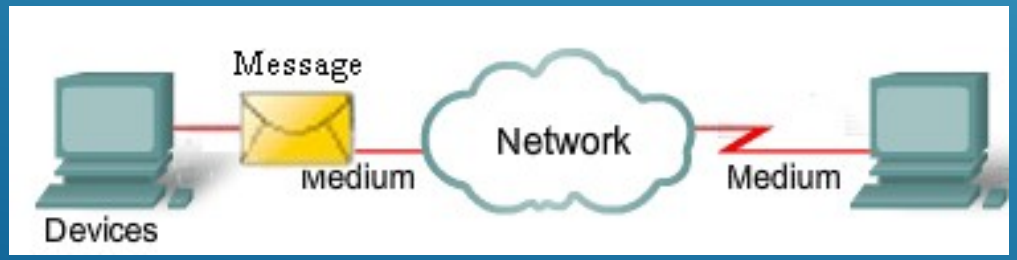



**Figure 2** Illustrates the Client as host and communicating with the Server.

# NETWORK COMPONENTS

## 2 Peer-to-Peer

Client and server software usually run on separate computers, but it is also possible for one computer to be used for both roles at the same time. In small businesses and homes, many computers function as the servers and clients on the network. This type of network is called a peer-to-peer network.

In the figure, the print-sharing PC has a Universal Serial Bus (USB) connection to the printer and a network connection, using a network interface card (NIC), to the file-sharing PC.

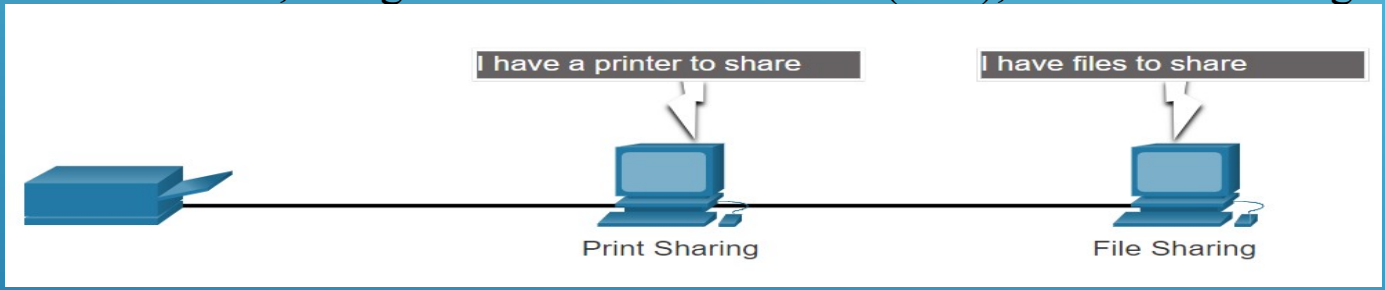



**Figure 3** Shows the peer-to-peer communication.

**Table 1** Shows the advantages and disadvantages of a Peer-to-Peer setup

| Advantages | Disadvantages |
|---|---|
| Easy to set up | No centralized administration |
| Less complex | Not a secure |
| Lower cost as network devices and dedicated servers are not essential | Not scalable |
| Used for a simple task like transfer files and sharing printers | All devices may act as both clients and servers which can slow their performance. |

# NETWORK COMPONENTS

### 3 End devices

The network devices that people are most familiar with are end devices. To distinguish one end device from another, each end device on a network has an address. When an end device initiates communication, it uses the address of the destination end device to specify where to deliver the message. An end device is either the source or destination of a message transmitted over the network.

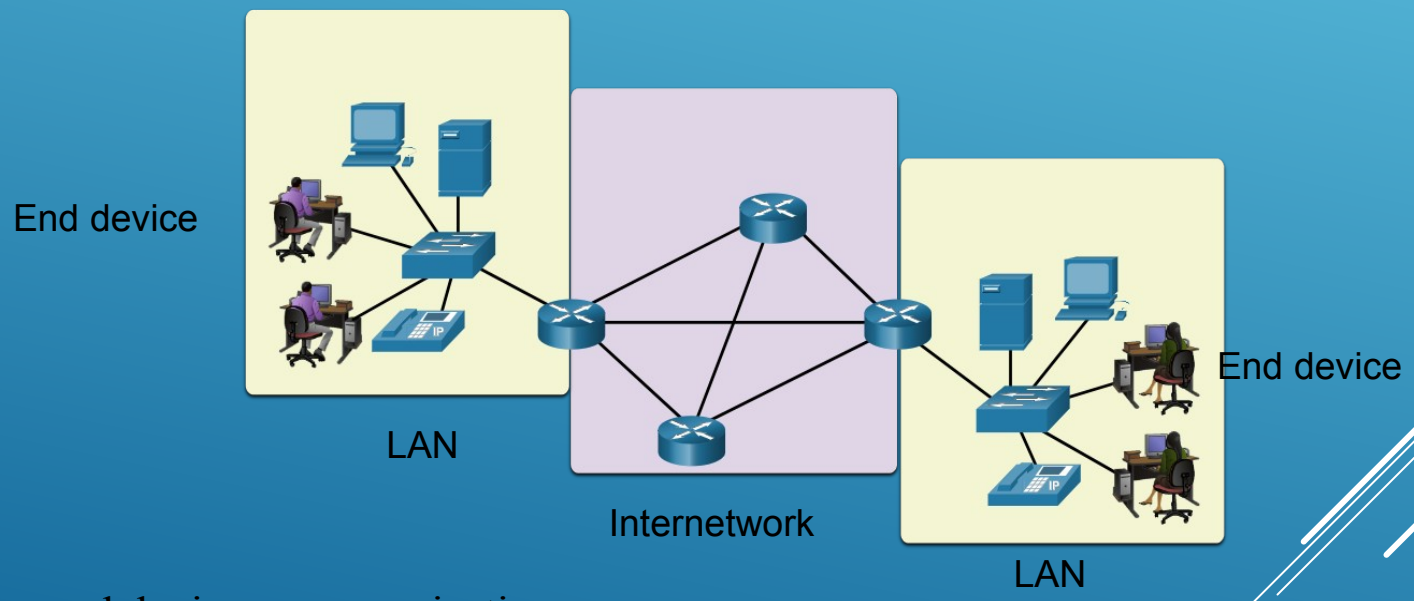



**Figure 4** shows end devices communication

Data originates from the end device, flows through the network, and reaches an end device.

# NETWORK COMPONENTS

### 4 Intermediary devices

Intermediary devices connect the individual end devices to the network. They can connect multiple individual networks to form an internetwork. These intermediary devices provide connectivity and ensure that data flows across the network. Intermediary devices use the destination end device address, in conjunction with information about the network interconnections, to determine the path that messages should take through the network. Examples of the more common intermediary devices and a list of functions are shown in the figure.

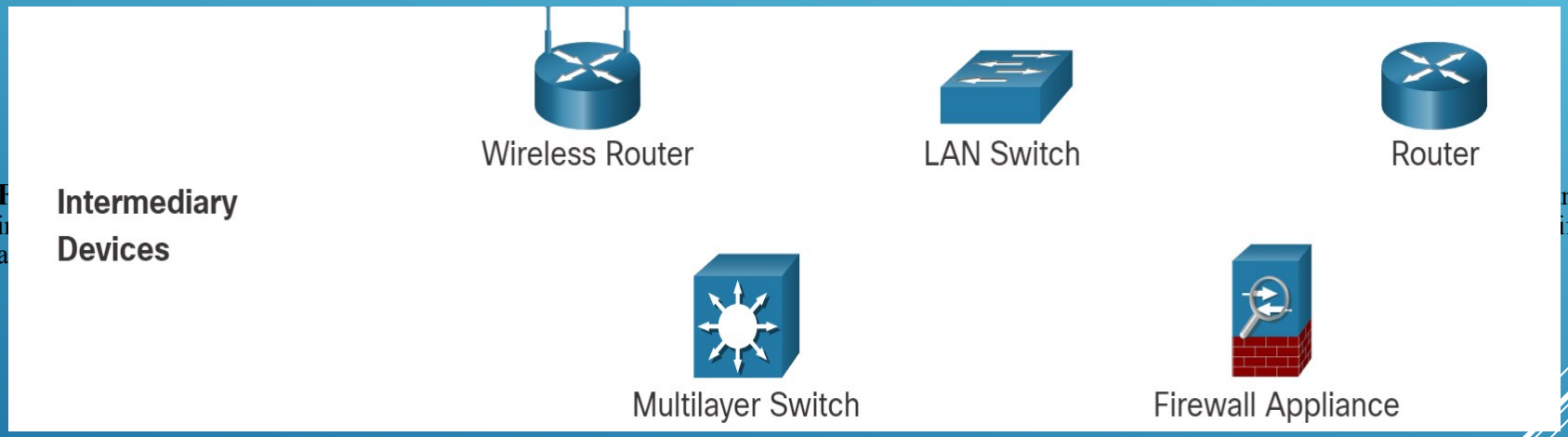



**Figure 5** Shows the Intermediary devices used in the network.

# NETWORK COMPONENTS

## 5 Network Media

Communication transmits across a network of media. The media provides the channel over which the message travels from source to destination. Modern networks primarily use three types of media to interconnect devices, as shown in the figure:

- **Metal wires within cables** - Data is encoded into electrical impulses.
- **Glass or plastic fibers within cables (fiber-optic cable)** - Data is encoded into pulses of light.
- **Wireless transmission** - Data is encoded via modulation of specific frequencies of electromagnetic waves.

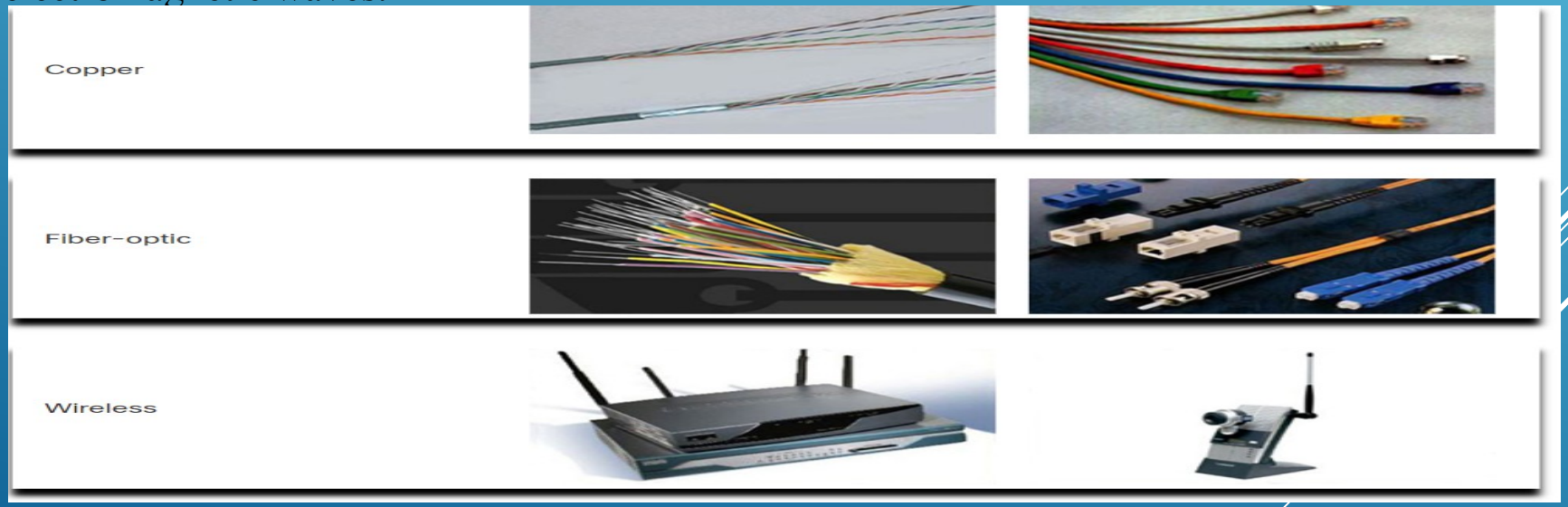



**Figure 6** Various ways of communication transmit across a network of media.

**Criteria for choosing:** max. distance, environment, amount of data, cost of media, and installation.

# NETWORK REPRESENTATION

In addition to these network components, specialized terminology is used to describe how each of these devices and media connect:

- **Network Interface Card (NIC)** - A NIC physically connects the end device to the network.
- **Physical Port** - A connector or outlet on a networking device where the media connects to an end device or another networking device.
- **Interface** - Specialized ports on a networking device that connect to individual networks. Because routers connect networks, the ports on a router are referred to as network interfaces.

**Note**: The terms port and interface are often used interchangeably.

# Different Types of Computer Networks

The classification of networks in computers can be done according to their

i)Size: expressed by the geographic area and number of computers

ii)Purpose: devices housed in a single room to millions of devices spread across the world.

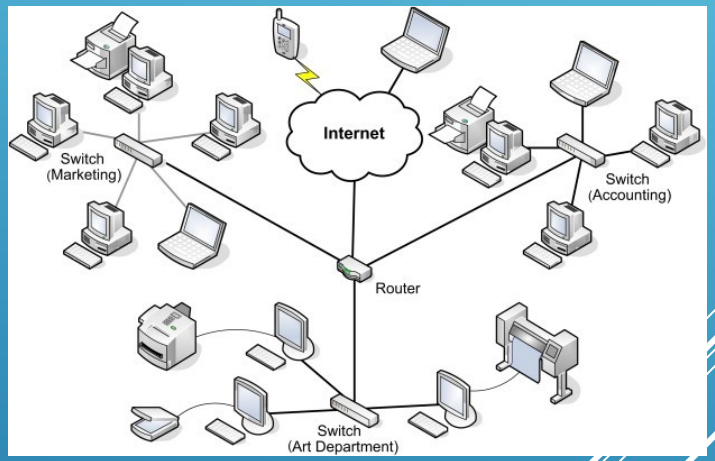



**Figure 7** Generic computer network connections

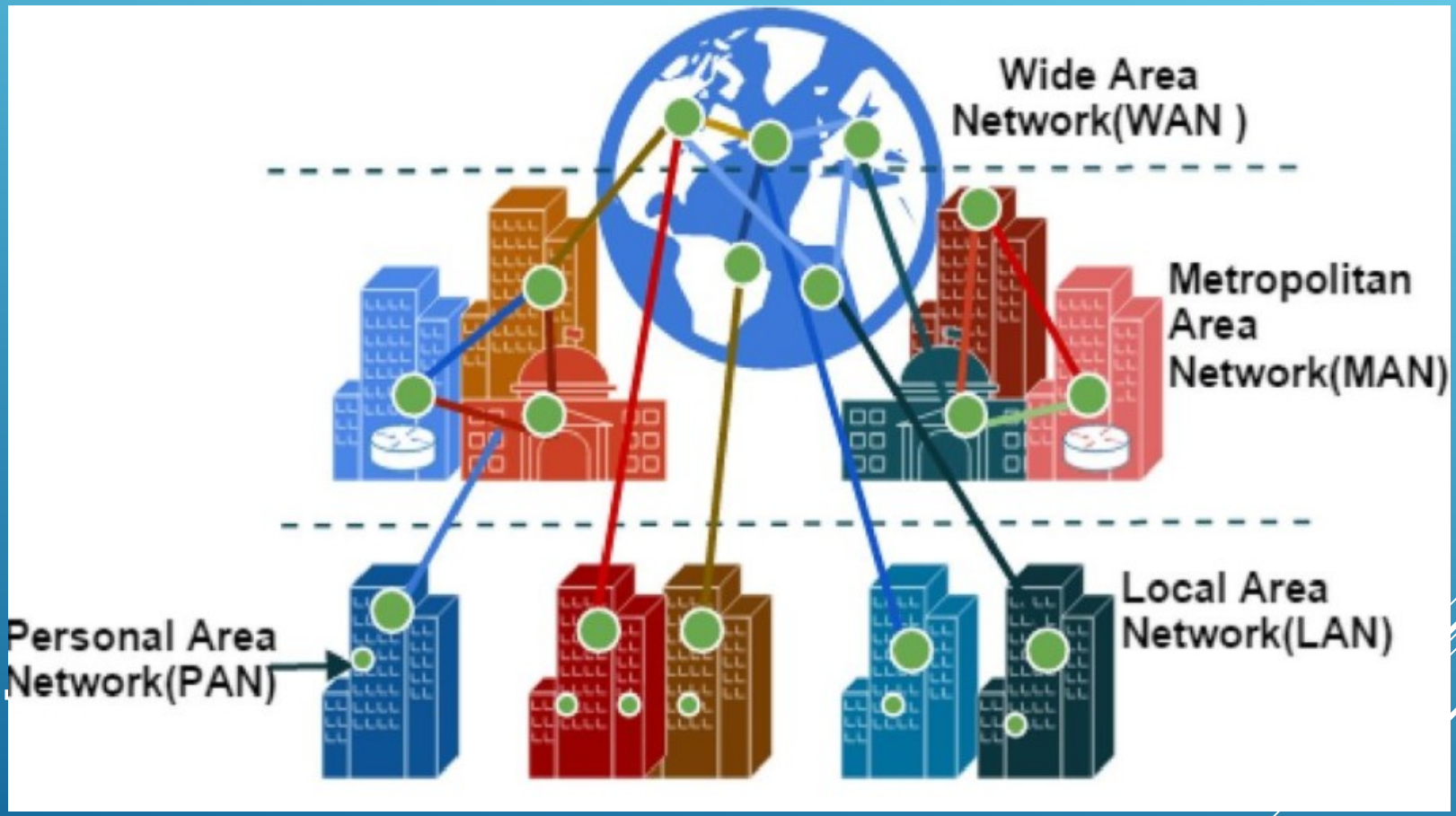



**Figure 8** Types of computer networks

# PERSONAL AREA NETWORK

**PAN** (Personal Area Network) is a computer network formed around a person.

**Characteristics of PAN**

Below are the main characteristics of PAN:

•It is mostly personal devices network equipped within a limited area.

•Appliances use for PAN: cordless mice, keyboards, and Bluetooth systems.

**Figure 9** Shows some of the PAN (Personal Area Network) appliances

# LOCAL AREA NETWORK

A group of computer and peripheral devices which are connected in a limited area such as school, laboratory, home, and office building.

The simplest type of LAN network is to connect computers and a printer in someone's home or office.

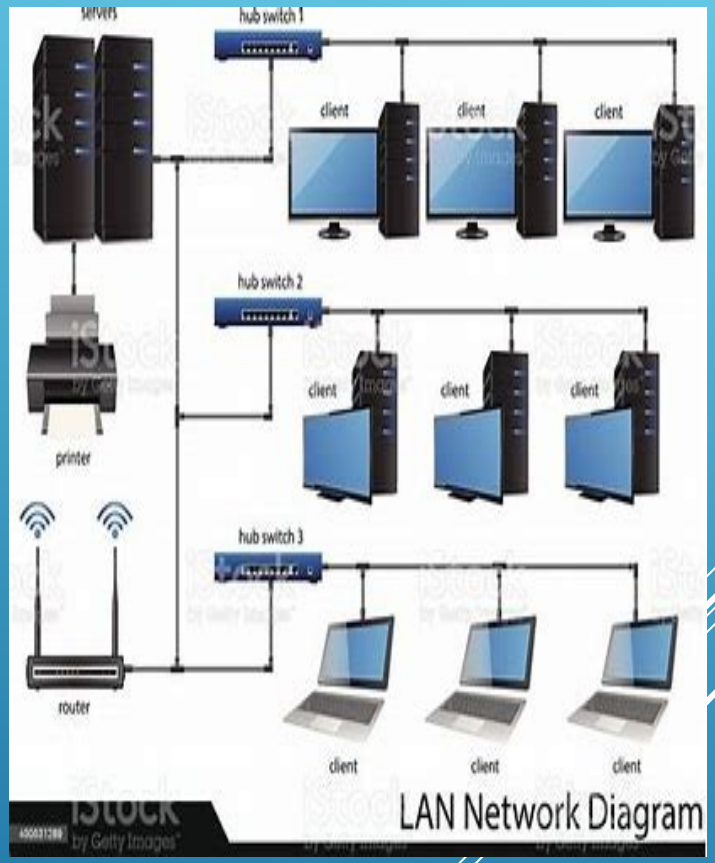



**Figure 10** The simplest type of LAN network

# LOCAL AREA NETWORK

**Characteristics of LAN**

Here are the important characteristics of a LAN network:

•It is a private network, so an outside regulatory body never controls it.

•LAN operates at a relatively higher speed compared to other WAN systems.

**Figure 11** Generic workflow of the simplest type of LAN network

# ADVANTAGES AND DISADVANTAGES OF LAN

| Advantages | Disadvantages |
|---|---|
| ✓Cheaper costs.<br>✓Easier to control.<br>✓Not much is required in setting up security to dispel hackers.<br>✓Workstations can share peripheral devices.<br>✓Do not necessarily need their own hard disk or CD-ROM drives.<br>✓User can save their work centrally on the network's file server.<br>✓Users can communicate with each other and transfer data between workstations very easily.<br>✓One copy of each application package can be loaded onto the file and shared by all users. | ✓Covers a small geographical region<br>✓Limitation in exchanging information.<br>✓Consumes a lot of power (it has to be on all the time for a good LAN).<br>✓Special security measures are needed to stop users from using programs and data that they should not have access to.<br>✓Networks are difficult to set up and need to be maintained by skilled technicians.<br>✓If the file server develops a serious fault, all the users are affected, rather than just one user in the case of a stand-alone machine. |

# MAN: METROPOLITAN AREA NETWORK

MAN stands for Metropolitan Area Network. It is a computer network that connects number of LANs to form larger network, so that the computer resources can be shared. This type of network covers larger area than a LAN but smaller than the area covered by a WAN which is designed to extend over the entire city. MAN is specially designed to provide high-speed connectivity to the users in which the speed ranges in terms of Mbps. The architecture of MAN is quite complicated hence, it is hard to design and maintain.

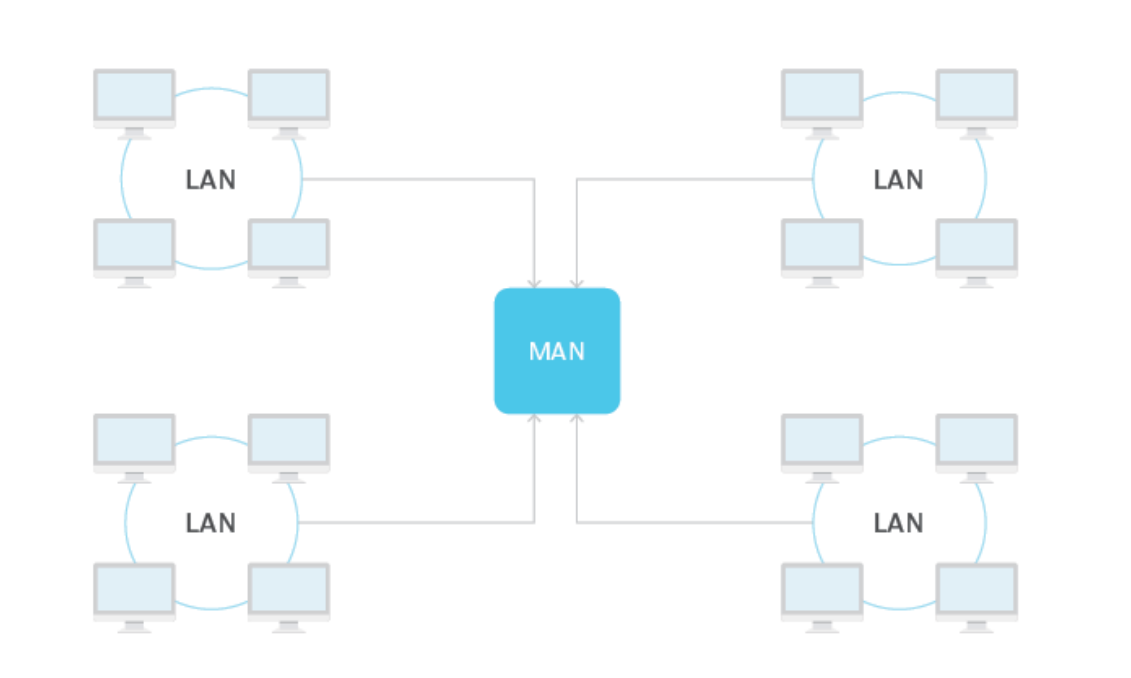



**Figure 12** The simplest type of MAN network

# WAN: WIDE AREA NETWORK

WAN is another  important computer network that which is spread across a large geographical area.

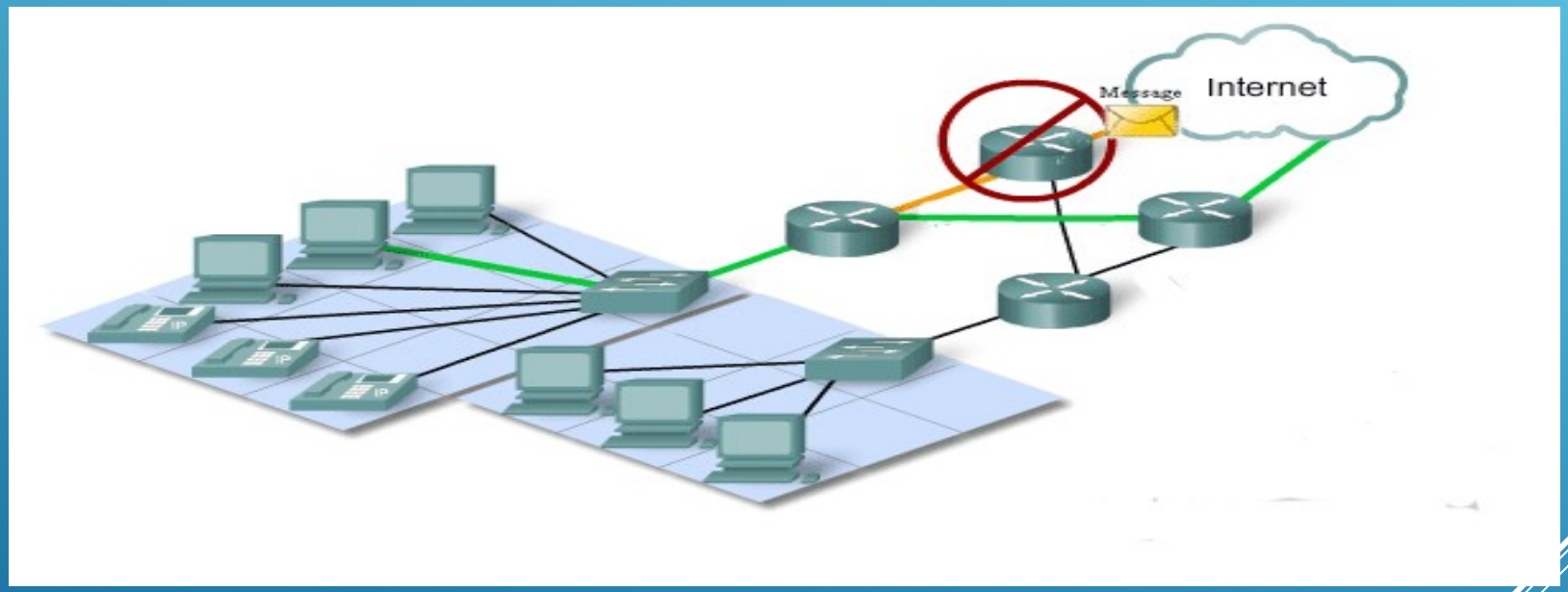


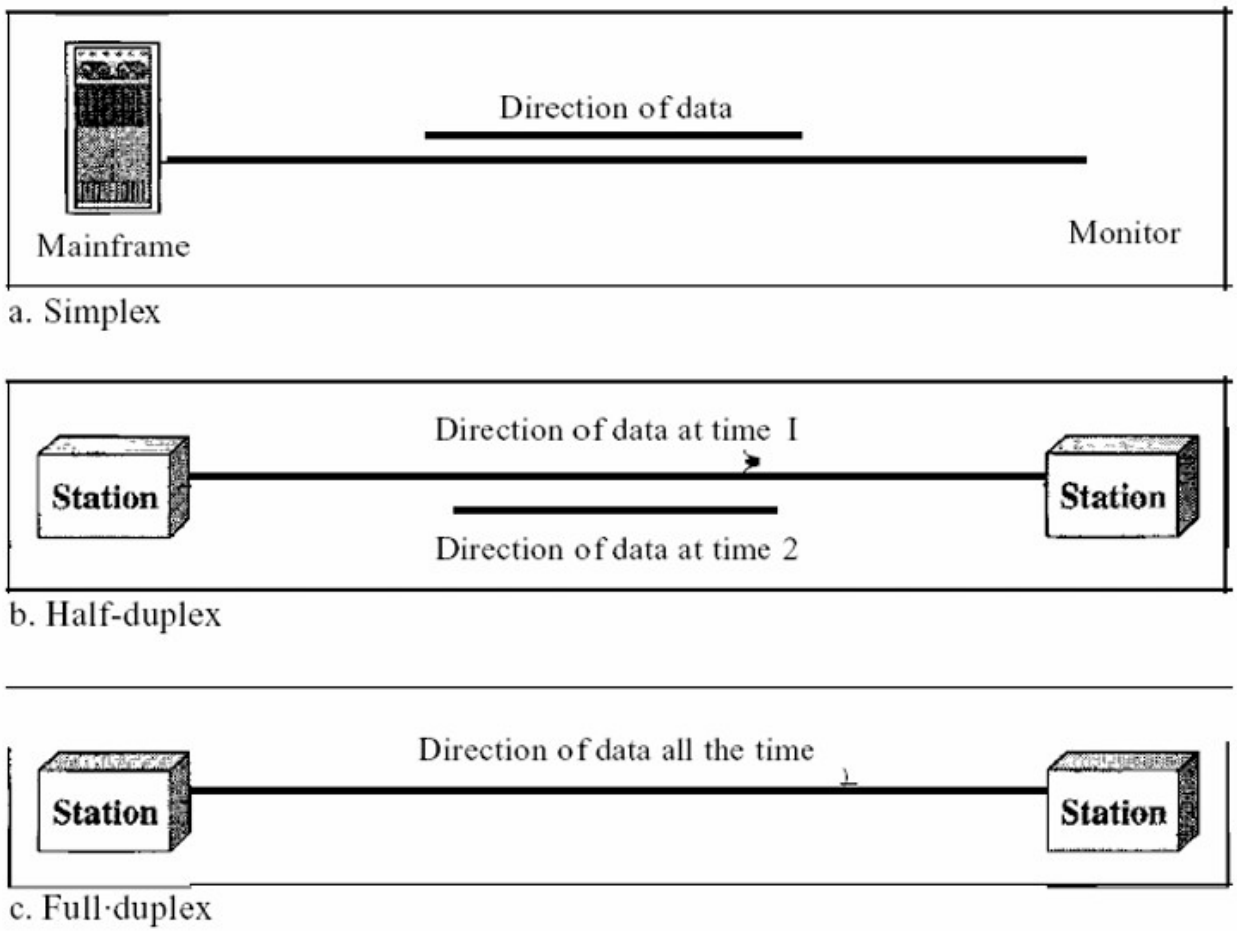

**Figure 12** The simplest type of WAN network

# Types of wireless networks

| | Wireless LAN (WLAN) | Wireless MAN (WMAN) | Wireless PAN (WPAN) | Wireless WAN (WWAN) |
|---|---|---|---|---|
| TYPE OF NETWORK | Local area network | Metropolitan area network | Personal area network | Wide area network |
| GOAL | Provide internet access within a building or limited outdoor area | Provide access outside office and home networks, typically regional | Transmit signals between devices in limited areas, typically 10 meters | Provide access outside the range of WLANs and WMANs |
| CONNECTIVITY | Radio waves, IEEE 802.11 Wi-Fi | IEEE 802.16 WiMax | Bluetooth, Zigbee and infrared | Cellular, LTE, 5G |

# DATA COMMUNICATION: DATA FLOW

Communication between two devices can be simplex, half-duplex, or full-duplex as shown in Figure.



# DATA COMMUNICATION: DATA FLOW

**Simplex** In simplex mode, the communication is unidirectional, as on a one way street. Only one of the two devices on a link can transmit; the other can only receive (Figure a). Keyboards and traditional monitors are examples of simplex devices.

**Half-Duplex**

In half-duplex mode, each station can both transmit and receive, but not at the same time. When one device is sending, the other can only receive, and vice versa (Figure b). Walkie-talkies and CB (citizens band) radios are both half duplex systems.

**Full-Duplex**

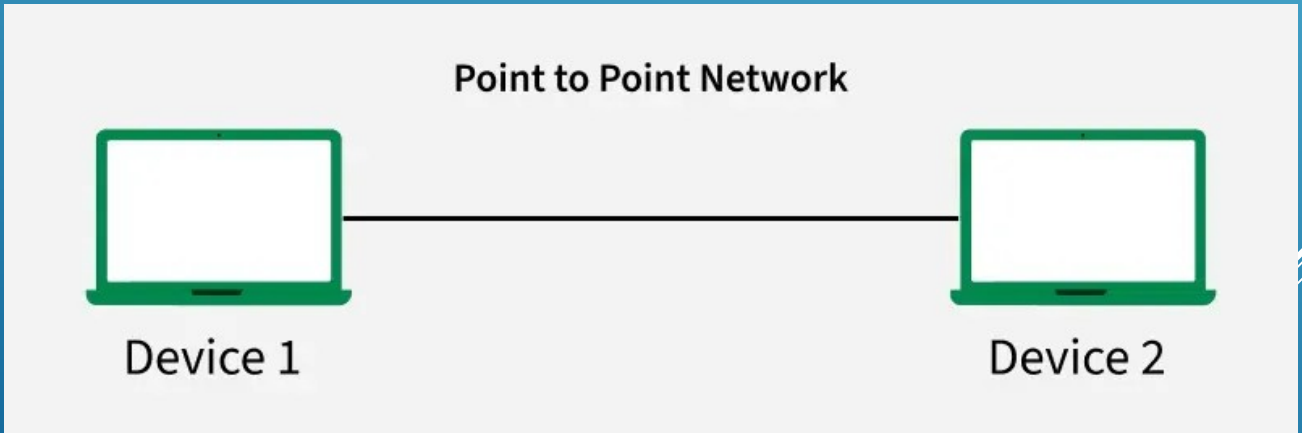
In full-duplex, both stations can transmit and receive simultaneously (Figure c). One common example of full-duplex communication is the telephone network. When two people are communicating by a telephone line, both can talk and listen at the same time. The full-duplex mode is used when communication in both directions is required all the time.

# BASED ON TYPES OF COMMUNICATION

Networks can also be divided based on the types of communication they use:

**1. Point-to-Point**

A Point-to-Point network creates a direct connection between two devices (e.g., computer and printer). Data is transmitted without intermediaries, using either wired (cables) or wireless (infrared, radio) links. It is simple, dedicated, and commonly used for two-device communication.



# BASED ON TYPES OF COMMUNICATION

**1. Point-to-Point**

**Advantages:**

- Provides a dedicated and direct connection, ensuring reliable communication between two devices.
- Simple to set up and configure due to the limited number of devices involved.
- Offers low latency as there are no intermediate devices or complex routing.

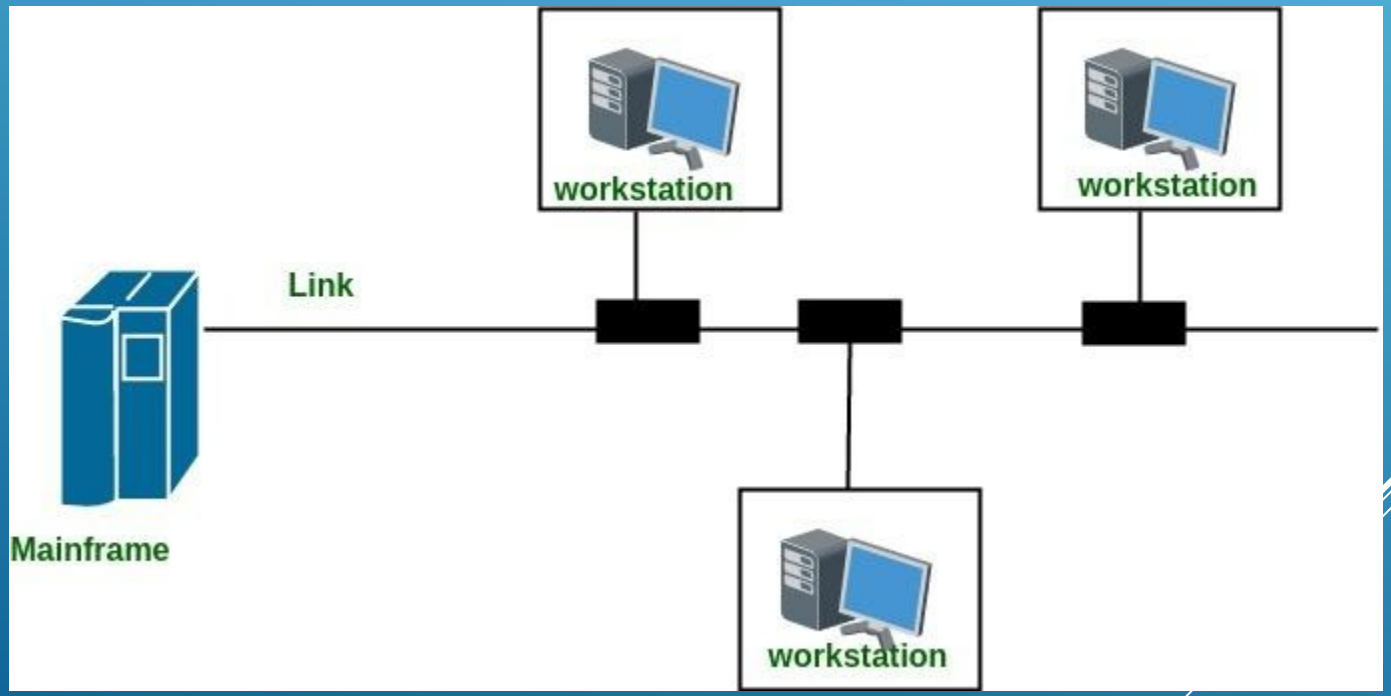
**Disadvantages:**

- Not scalable, as adding more devices requires additional dedicated links.
- May require physical cables for wired connections, which can limit flexibility.
- Lacks fault tolerance; if the direct link fails, communication is disrupted.

# BASED ON TYPES OF COMMUNICATION

**2. Multipoint Networks**

A multipoint network allows more than two devices to share a single communication link. The channel capacity is either shared at the same time (spatially) or in turns (temporally).



# BASED ON TYPES OF COMMUNICATION

**2. Multipoint Networks**

Advantages:

- Allows multiple devices to share a single communication channel, reducing the need for multiple dedicated links.
- More cost-effective than point-to-point networks for connecting multiple devices.
- Supports dynamic communication, where devices can join or leave the network easily.

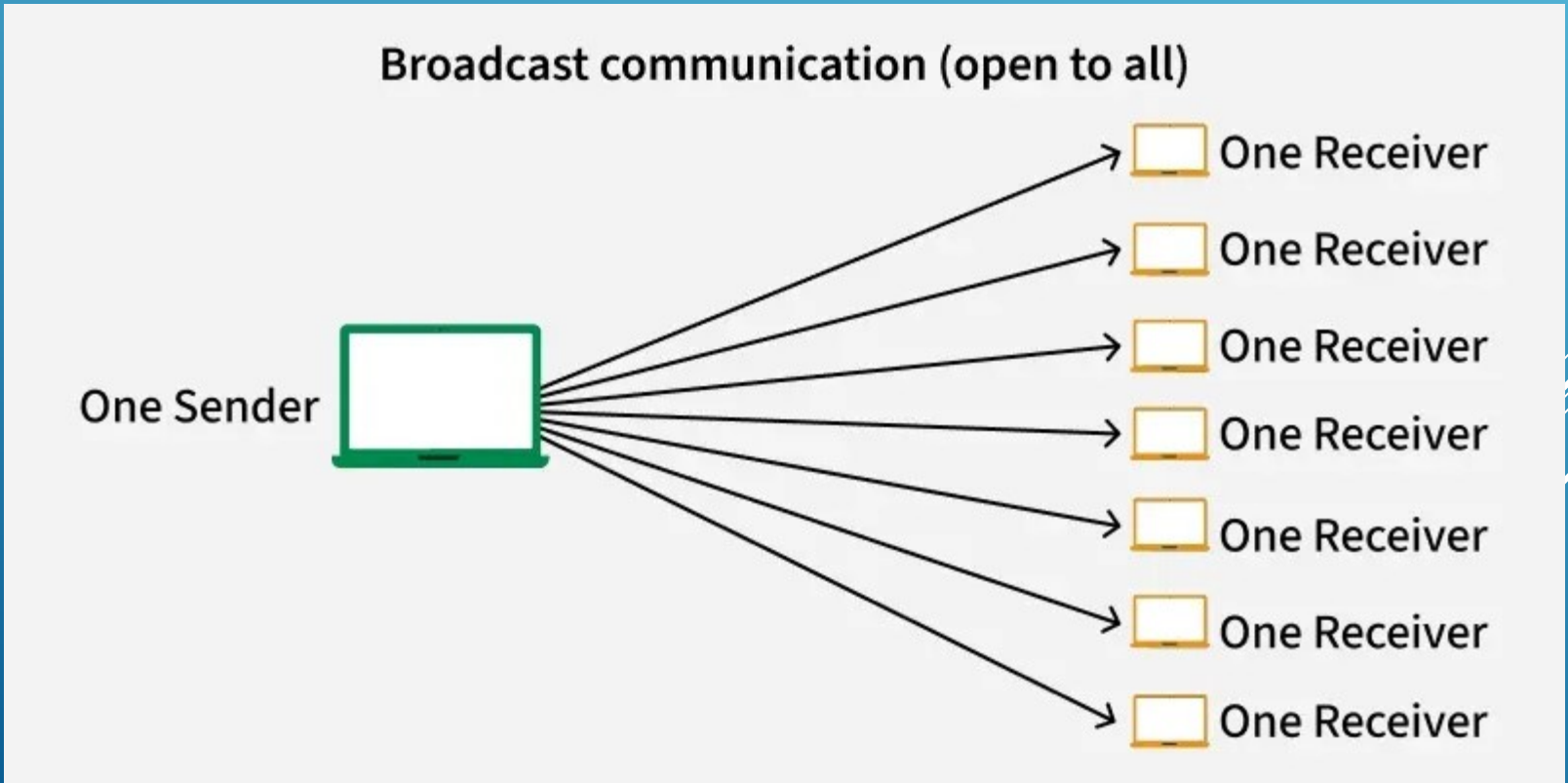
Disadvantages:

- Shared bandwidth can lead to congestion and reduced performance when many devices are active.
- Potential for interference or collisions in spatially shared connections.
- Security risks may increase as multiple devices access the same communication medium.

# BASED ON TYPES OF COMMUNICATION

**3. Broadcast Networks**

Broadcast networks use a single communication channel where one sender transmits data that can be received by multiple parties simultaneously. In this setup, data travels in one direction, from the sender to all potential receivers. A common example is a radio station, where the station broadcasts signals that can be picked up by any radio receiver within range. Another example is a television network broadcasting to multiple viewers.



# BASED ON TYPES OF COMMUNICATION

**3. Broadcast Networks**

**Advantages:**

- Efficient for distributing data to a large number of recipients simultaneously.
- Simple to implement for one-to-many communication scenarios.
- Reduces the need for individual connections to each receiver, saving resources.

**Disadvantages:**

- Data is sent in one direction, limiting two-way communication.
- No guarantee that all intended recipients receive the data reliably.
- Can lead to bandwidth wastage if not all receivers need the transmitted data.

# TOPOLOGIES

The Physical or Logical layout of a network is known as Topology of the network.

It can be divided into:
i) Physical Topology
ii)Logical Topology

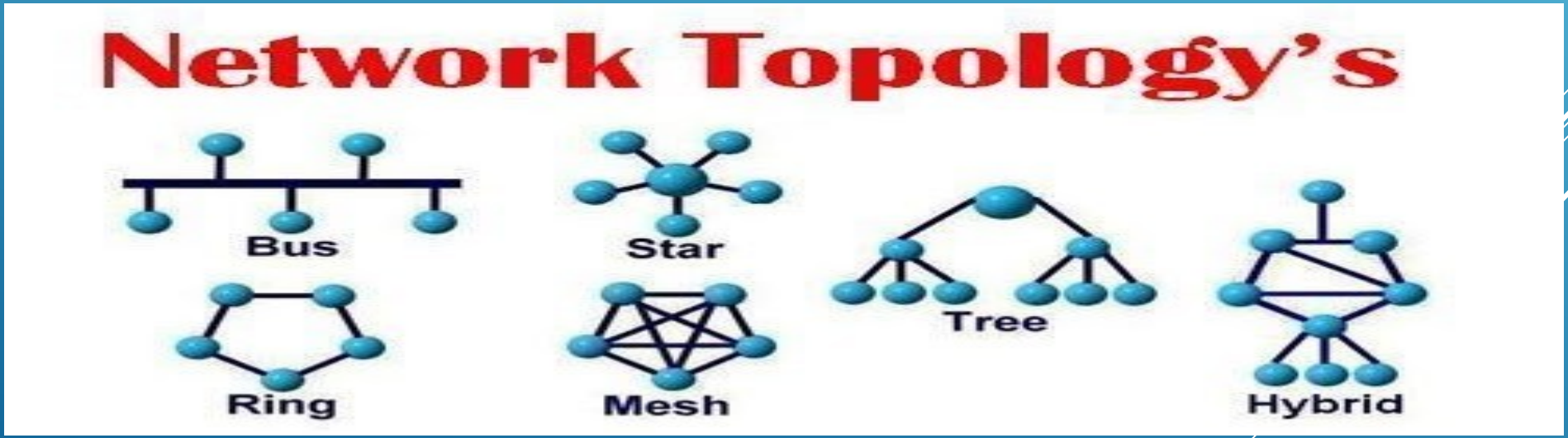



**Figure 14** Various network Topology's.

# TOPOLOGIES

## Physical vs. Logical Topology

- The actual layout of a network and its media is its Physical Topology
- The way in which the data access the medium and transmits packets is the Logical Topology
- A glance at a network is not always revealing. Cables emerging from a Hub does not make it necessarily a Star Topology – it may actually be a bus or a ring

# TOPOLOGIES

This example shows the department server, one admin, various classroom that are connected to each other by routers, ethernet, internet and various addresses. This depict the way data is accessed and transmit packets is example of logical topology.

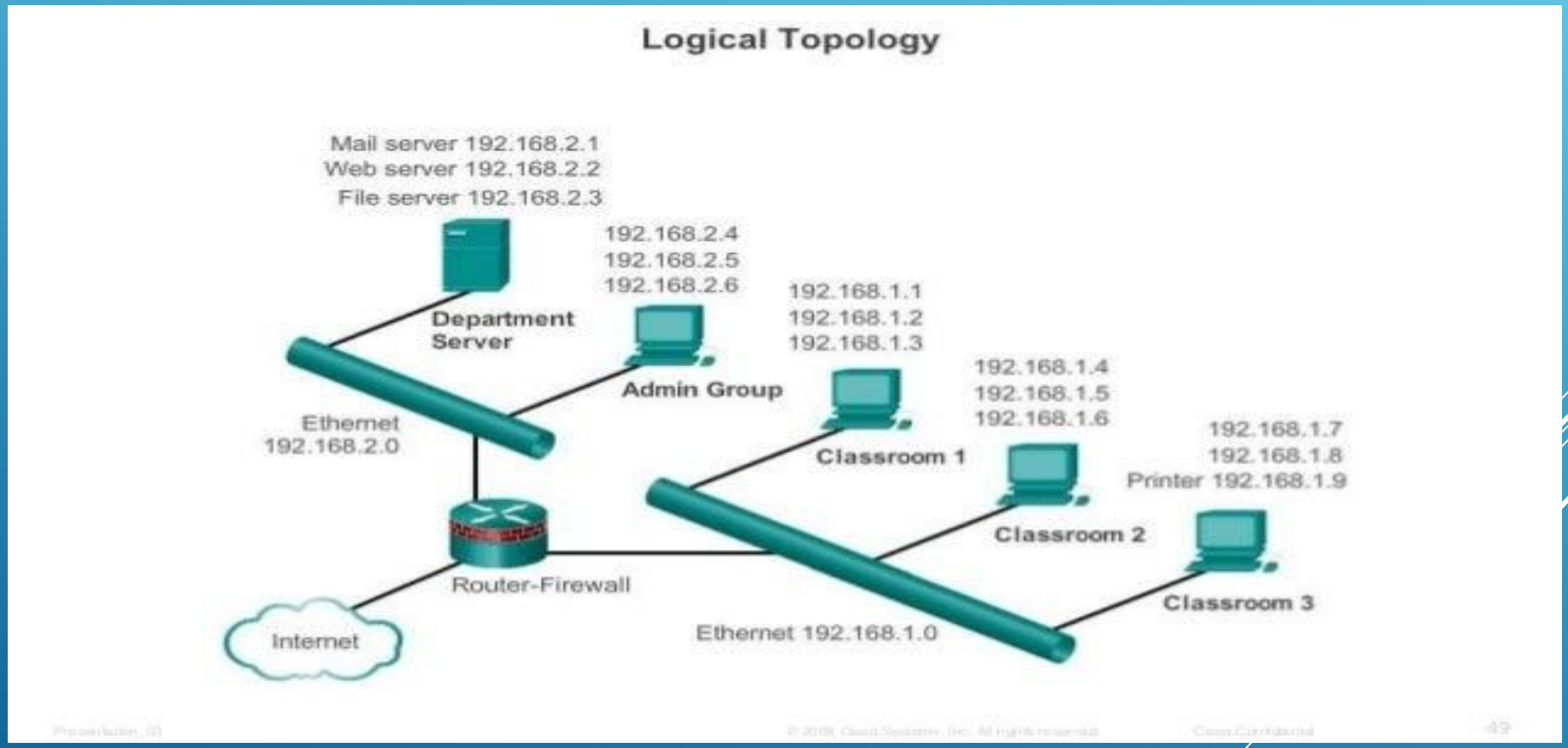



**Figure 15** Illustrates logical topology.

# MESH TOPOLOGY

In a mesh topology, every device is connected to another device via a particular channel.

Every device is connected with another via dedicated channels. These channels are known as links.

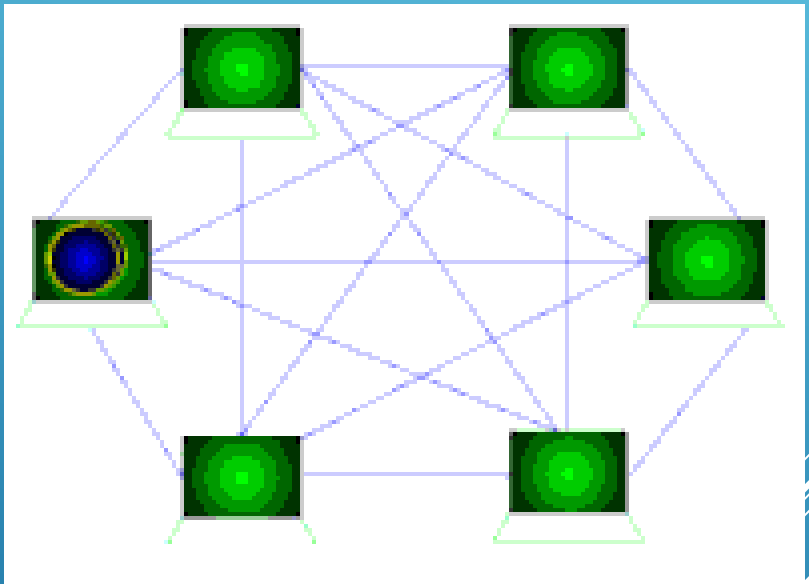


**Figure 16** Illustrates Mesh topology.

# Mesh Topology

- Suppose, N number of devices are connected with each other in a mesh topology, the total number of ports that are required by each device is N-1.
- In Figure 1, there are 5 devices connected to each other, hence the total number of ports required by each device is 4. Total number of ports required=N*(N-1).
- Suppose, N number of devices are connected with each other in a mesh topology, then the total number of dedicated links required to connect them is ${}^{N}C_2$ i.e. N(N-1)/2. In Figure 1, there are 5 devices connected to each other, hence the total number of links required is 5*4/2 = 10.

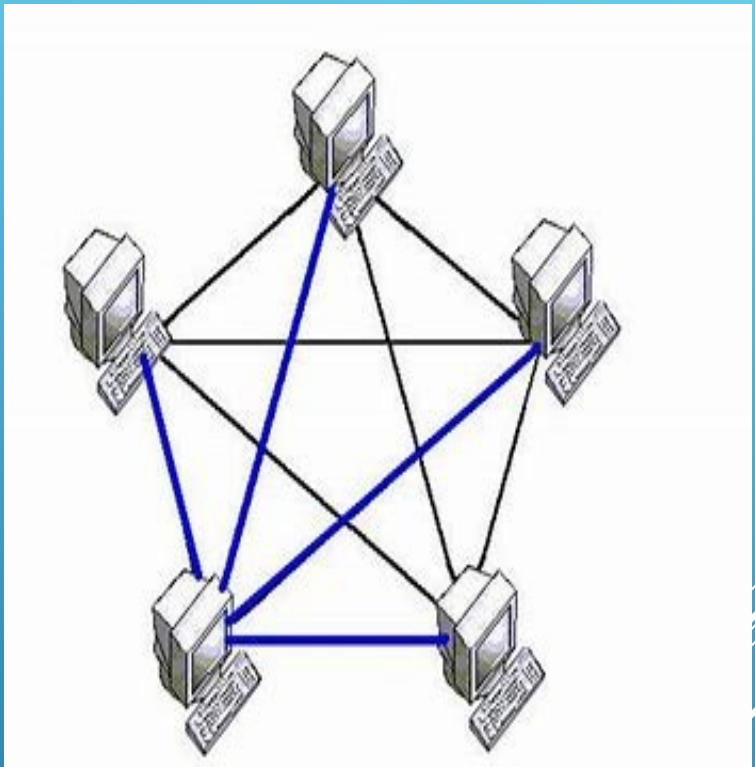


**Figure 17** Mesh topology with 5 computers connections.

# Star Topology

In star topology, all the devices are connected to a single networking device through a cable. This central device may be hub, switch, router  and all other nodes are connected to the central node.

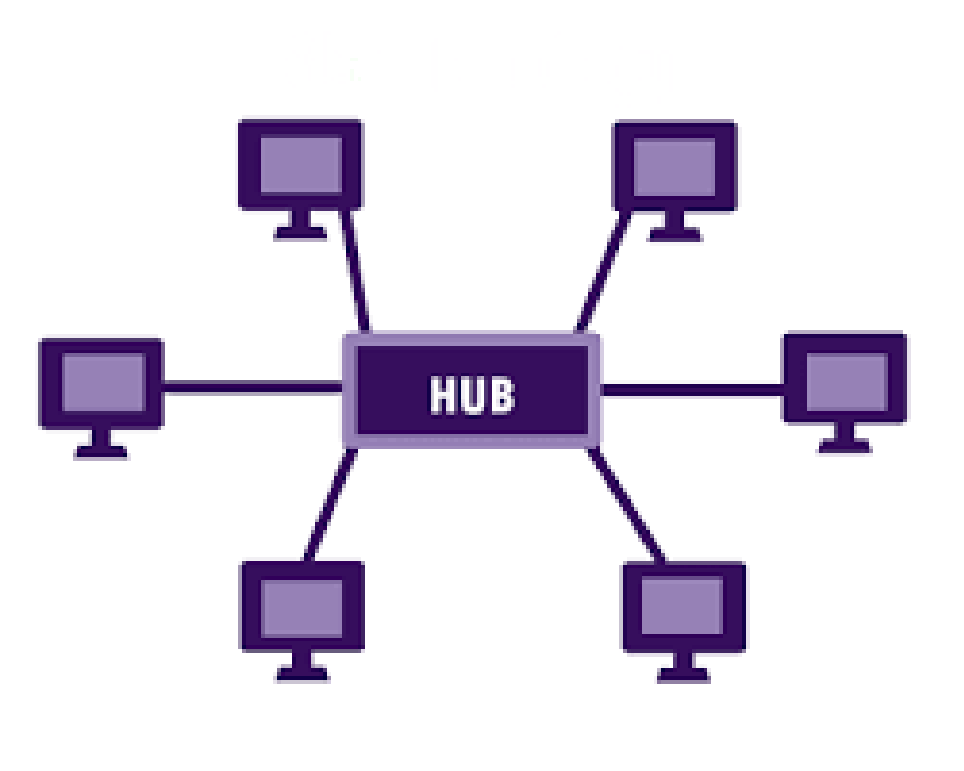



**Figure 18** Star topology, Hub at centre and rest all are PC's

# Star Topology

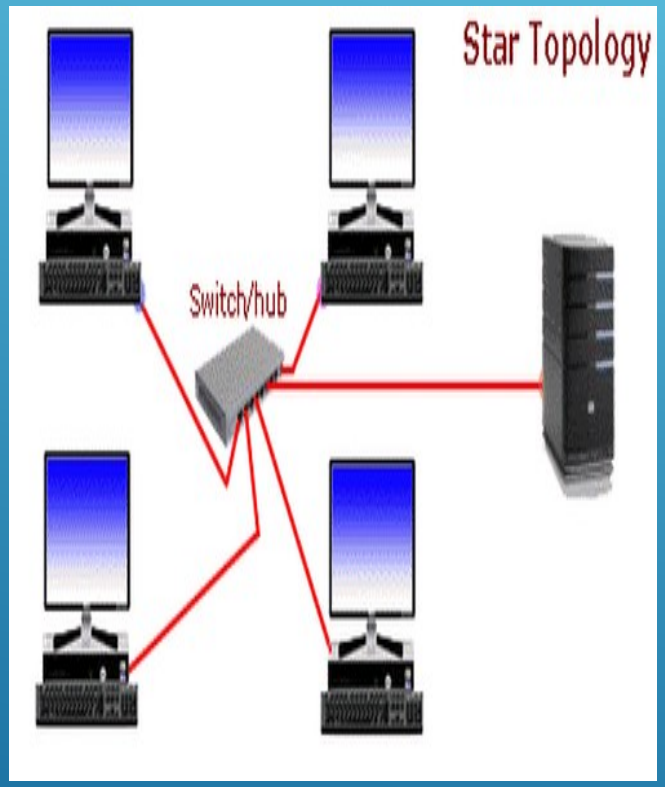



**Figure 19** Workflow of Star topology

**Advantages of this topology :**

•If N devices are connected in a star topology, then the number of cables required to connect them is N. So, it is easy to set up.

•Each device requires only 1 port i.e. to connect to the hub, therefore the total number of ports required is N.

**Problems with this topology :**

•If the concentrator (hub) on which the whole topology relies fails, the whole system will crash down.

•The cost of installation is high.

•Performance is based on the single concentrator i.e. hub.

# BUS TOPOLOGY

Bus topology is a network type in which every computer and network device is connected to a single cable. It transmits the data from one end to another in a single direction. No bi-directional feature is in bus topology.

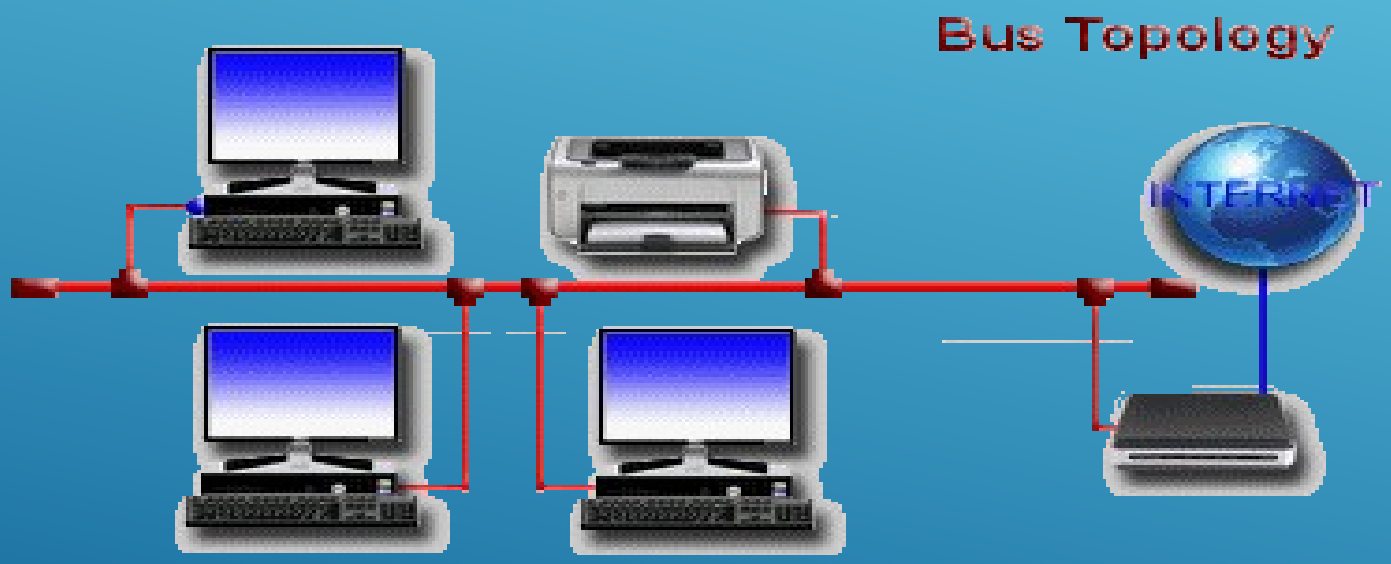



**Figure 20** Workflow of Bus topology

# BUS TOPOLOGY

**Advantages of this topology :**

•If N devices are connected in a bus topology, then the number of cables required to connect them is 1, which is known as backbone cable, and N drop lines are required.

•The cost of the cable is less as compared to other topologies, but it is used to build small networks.

**Problems with this topology :**

•If the common cable fails, then the whole system will crash down.

•If the network traffic is heavy, it increases collisions in the network. To avoid this, various protocols are used in the MAC layer known as Pure Aloha, Slotted Aloha, CSMA/CD, etc.

•Security is very low.

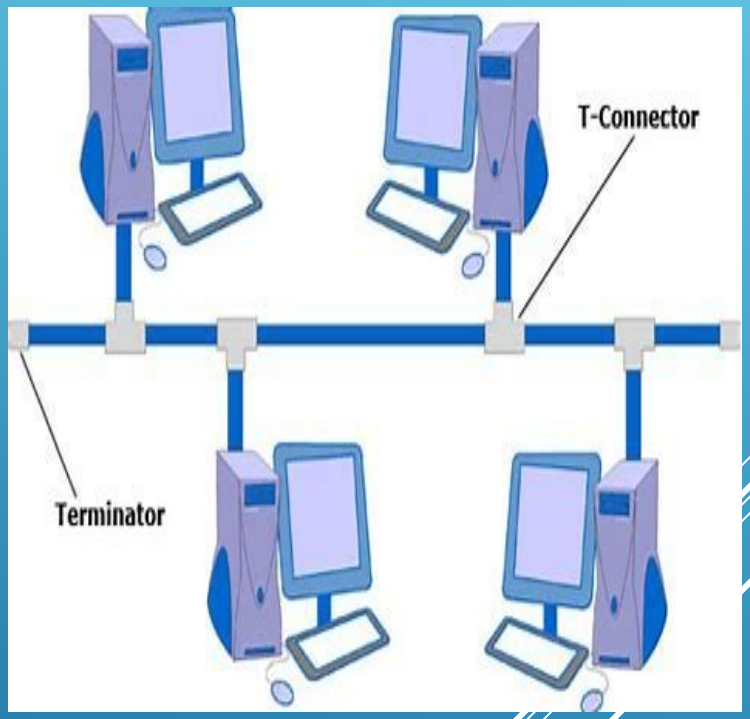



**Figure 21** Bus topology connection.

# RING TOPOLOGY

In this topology, it forms a ring connecting devices with exactly two neighboring devices.

Several repeaters are used for Ring topology with a large number of nodes, because if someone wants to send some data to the last node in the ring topology with 100 nodes, then the data will have to pass through 99 nodes to reach the 100th node. Hence to prevent data loss repeaters are used in the network.

The transmission is unidirectional, but it can be made bidirectional by having 2 connections between each Network Node, it is called Dual Ring Topology.

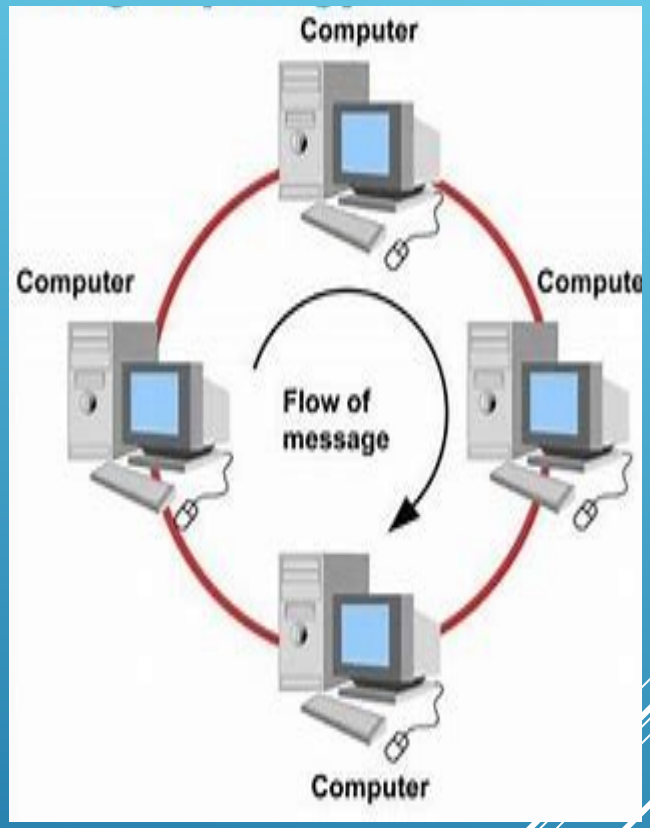



**Figure 22** Flow of message in Ring topology.

# RING TOPOLOGY

The following operations take place in ring topology are:

1. One station is known as a **monitor** station which takes all the responsibility to perform the operations.
2. To transmit the data, the station has to hold the token. After the transmission is done, the token is to be released for other stations to use.
3. When no station is transmitting the data, then the token will circulate in the ring.
4. There are two types of token release techniques: **Early token release** releases the token just after transmitting the data and **Delay token release** releases the token after the acknowledgment is received from the receiver.

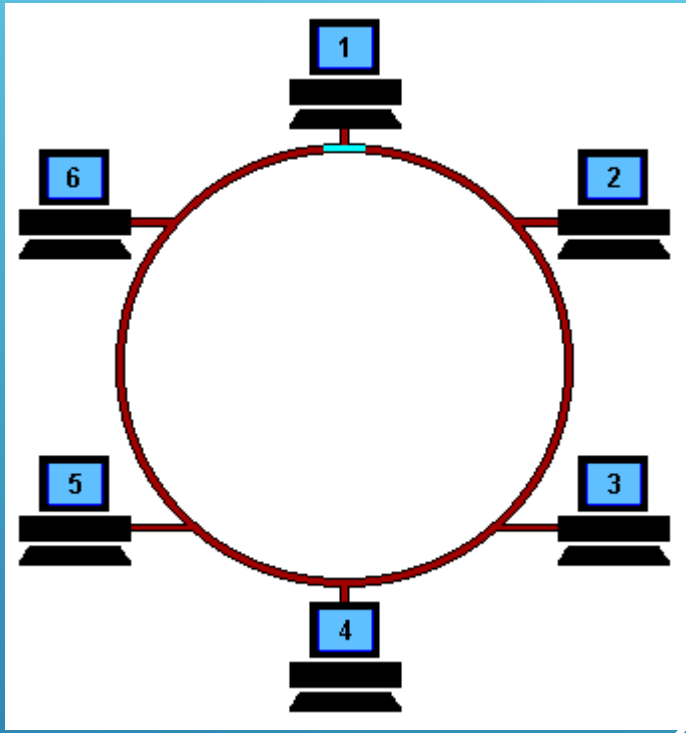



**Figure 23** Operations in Ring topology.

# RING TOPOLOGY

**Advantages of this topology :**

•The possibility of collision is minimum in this type of topology.

•Cheap to install and expand.

**Problems with this topology :**

•Troubleshooting is difficult in this topology.

•The addition of stations in between or removal of stations can disturb the whole topology.

•Less secure.

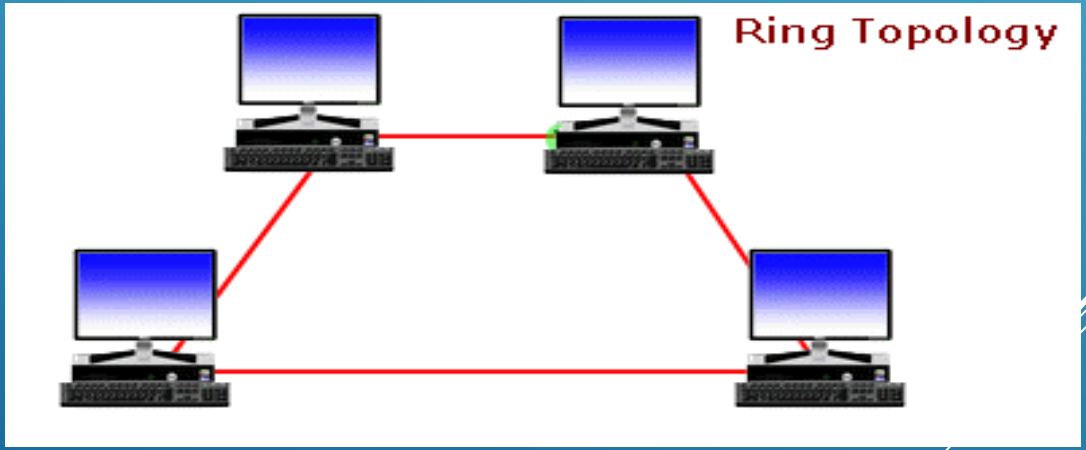



**Figure 24** Operations in Ring topology.

# TREE TOPOLOGY

This topology is the variation of Star topology. This topology has a hierarchical flow of data.

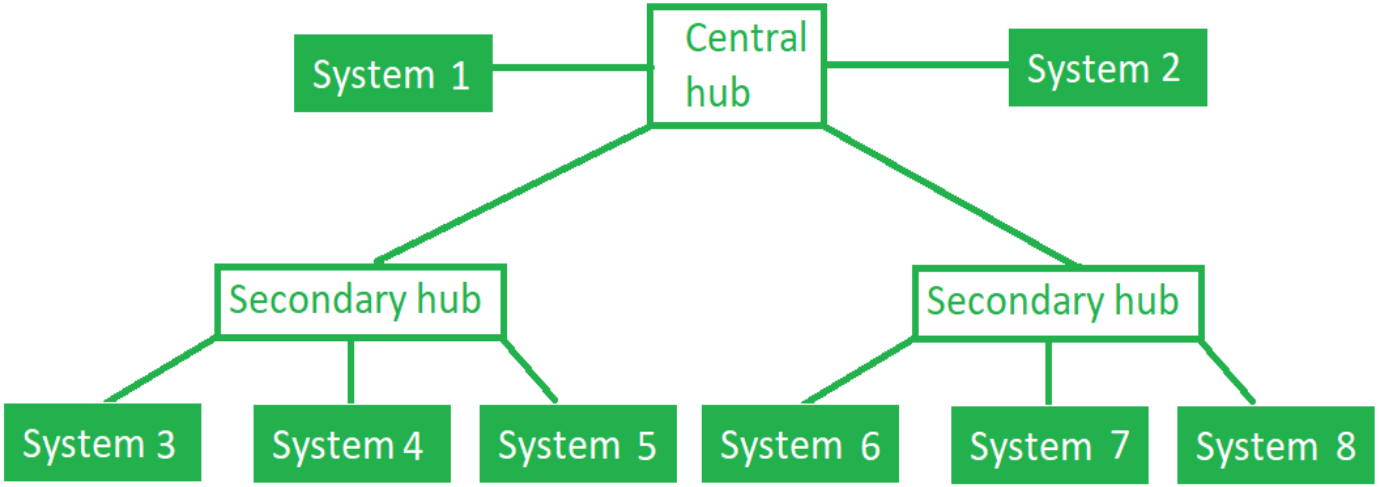



**Figure 25** Shows Tree topology's hierarchical flow of data.

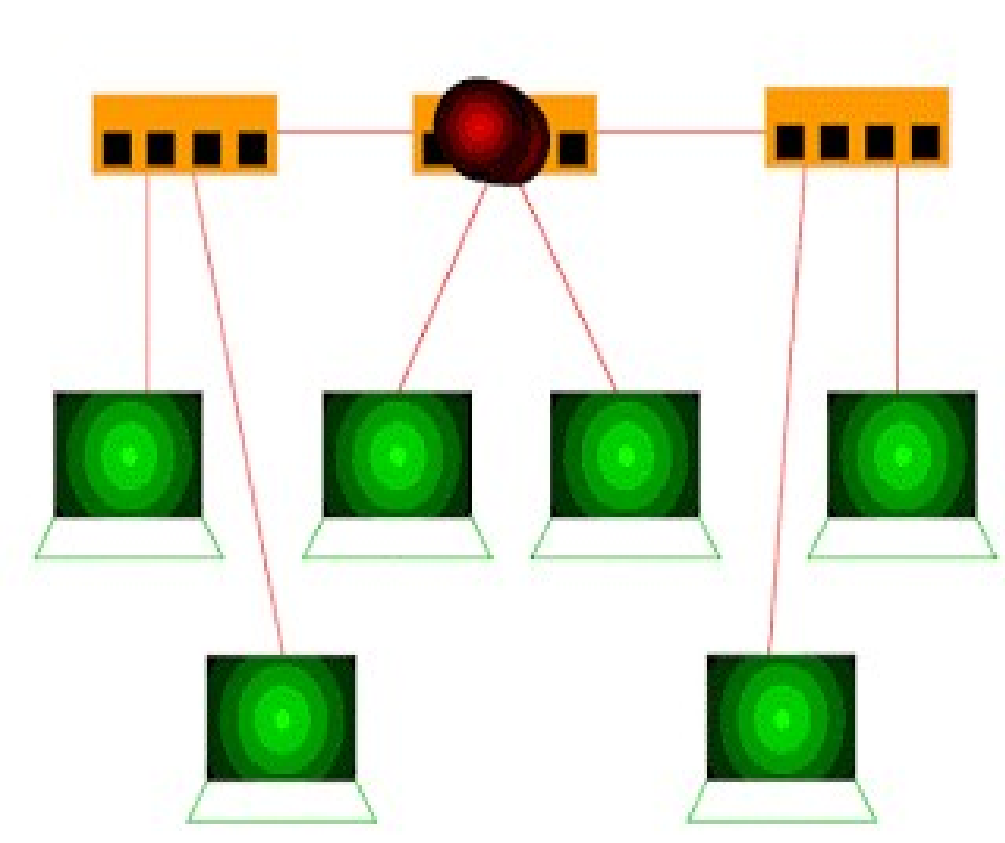


**Figure 26** Shows Tree topology's hierarchical flow of data.

# TREE TOPOLOGY

In this, the various secondary hubs are connected to the central hub which contains the repeater. In this data flow from top to bottom i.e. from the central hub to secondary and then to the devices or from bottom to top i.e. devices to the secondary hub and then to the central hub.

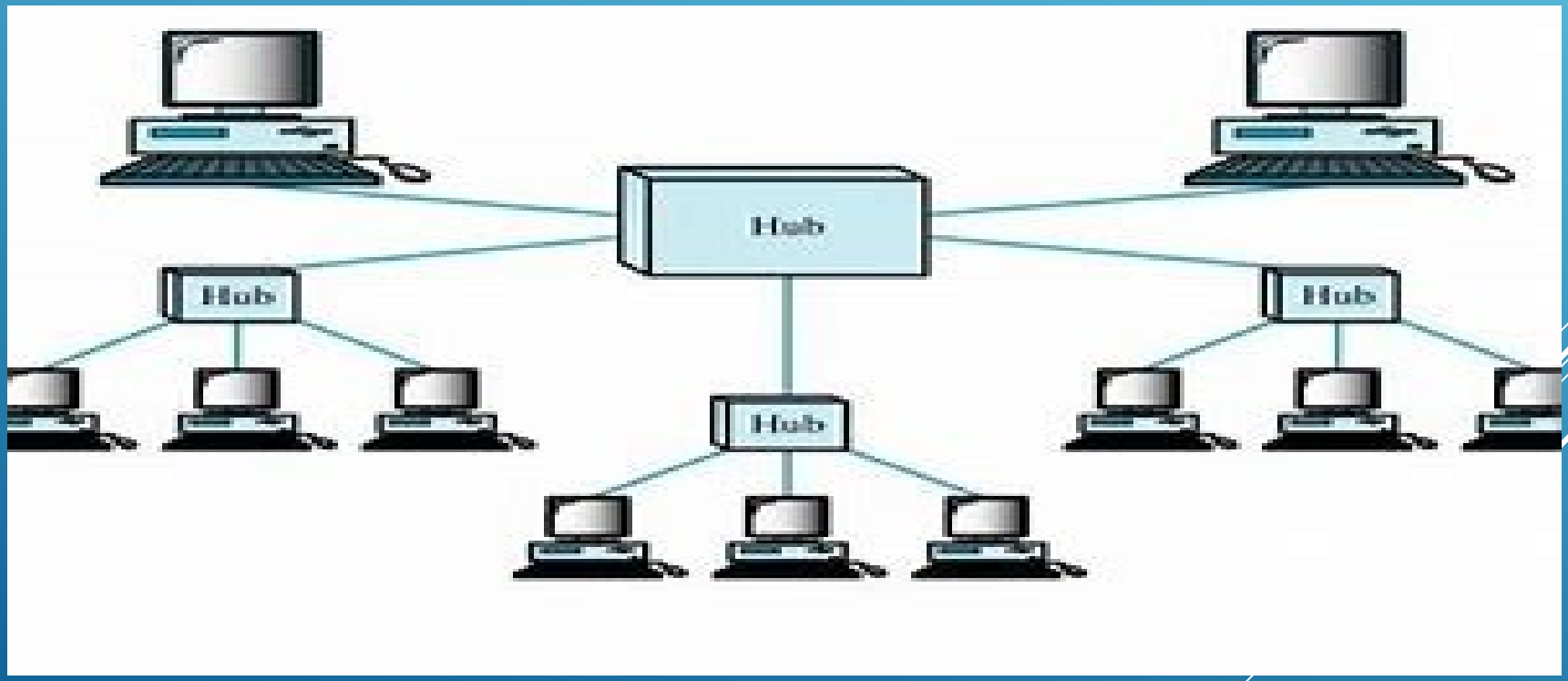



**Figure 26** Tree topology connected to the central and secondary hub.

# TREE TOPOLOGY

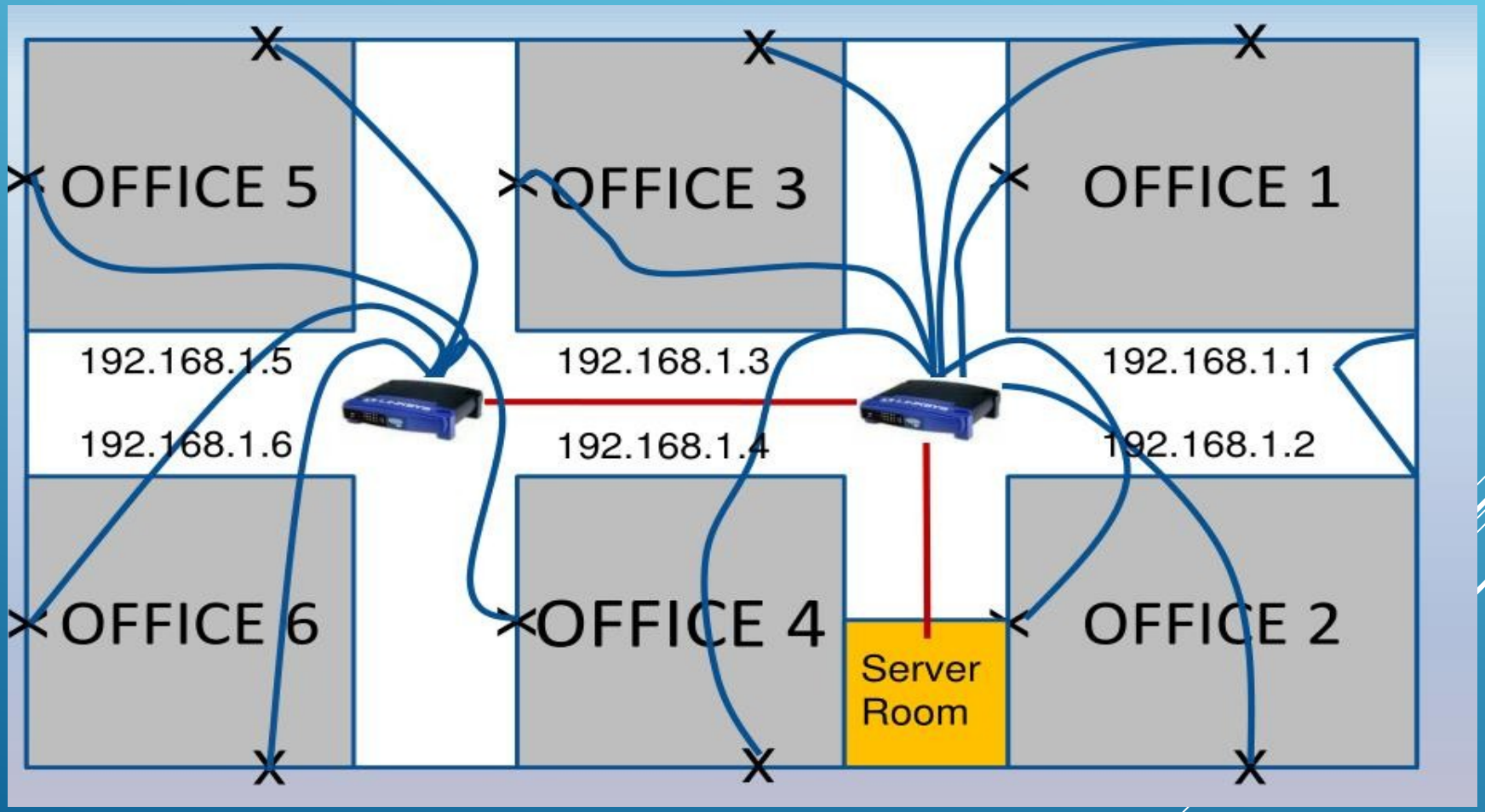



**Figure 27** Tree topology with specific addresses.

# TREE TOPOLOGY

**Advantages of this topology :**

•It allows more devices to be attached to a single central hub thus it decreases the distance that is traveled by the signal to come to the devices.

•It allows the network to get isolated and also prioritize from different computers.

**Problems with this topology :**

•If the central hub gets fails the entire system fails.

•The cost is high because of cabling.

- Cisco Packet Tracer is a powerful network simulation program that allows students to experiment with network behavior

# happy

# LEARNING